Lab Report: Training dynamics of deep models

1 Introduction

In this report we compare several models on standard benchmarks. The learning rate was set to 0.001 with the Adam optimizer. The dataset was split into 70 percent training data and the rest for testing.

2 Results

Figure 1 shows the test accuracy by dataset.

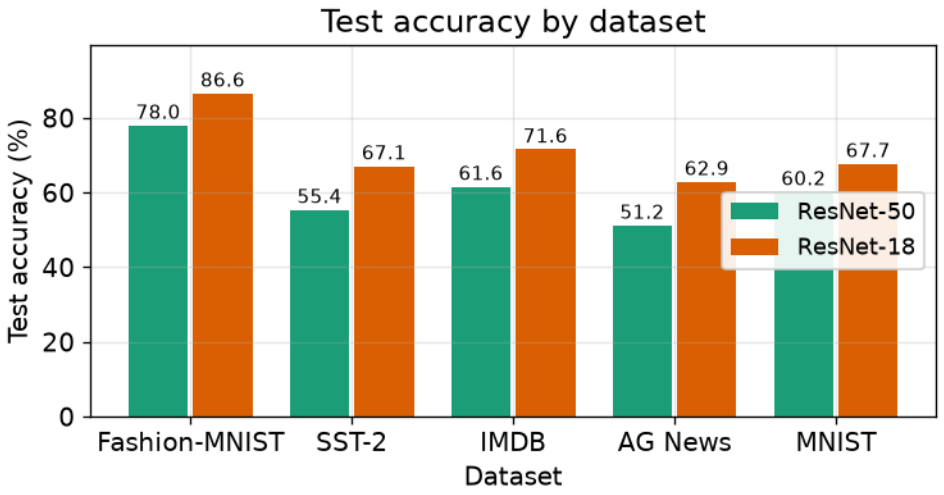


Figure 1: Test accuracy by dataset for the models in this experiment.

On IMDB, ResNet-50 obtains a test accuracy of over 60%. ResNet-18 reaches 67.7% test accuracy on MNIST (Figure 1).

Figure 2 shows the f1 score during training.

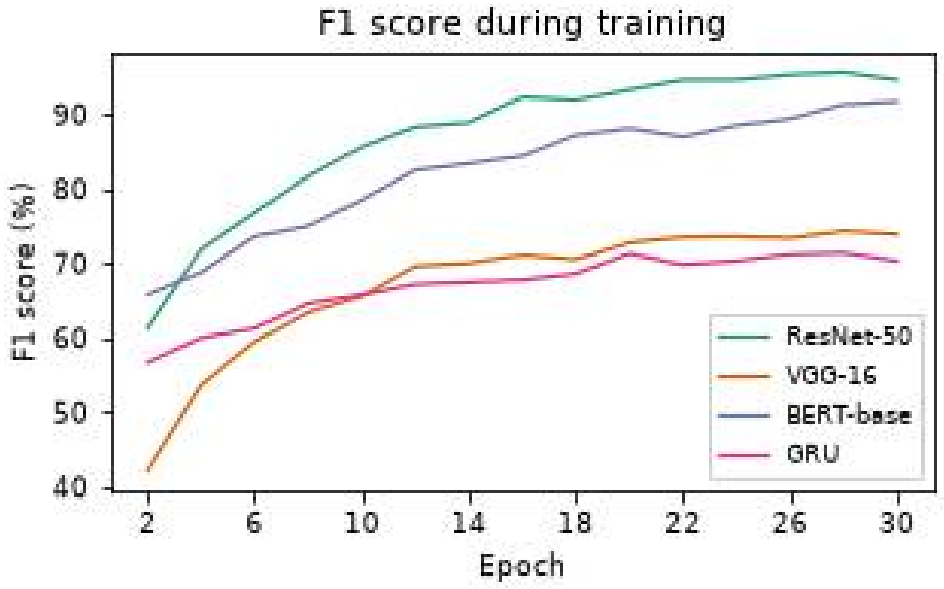


Figure 2: F1 score during training for the models in this experiment.

BERT-base attains a F1 score of 73.8% at epoch 8. As shown in Figure 2, at epoch 12, ResNet-50 is 21.6 points better than VGG-16.

Figure 3 shows the f1 score by dataset.

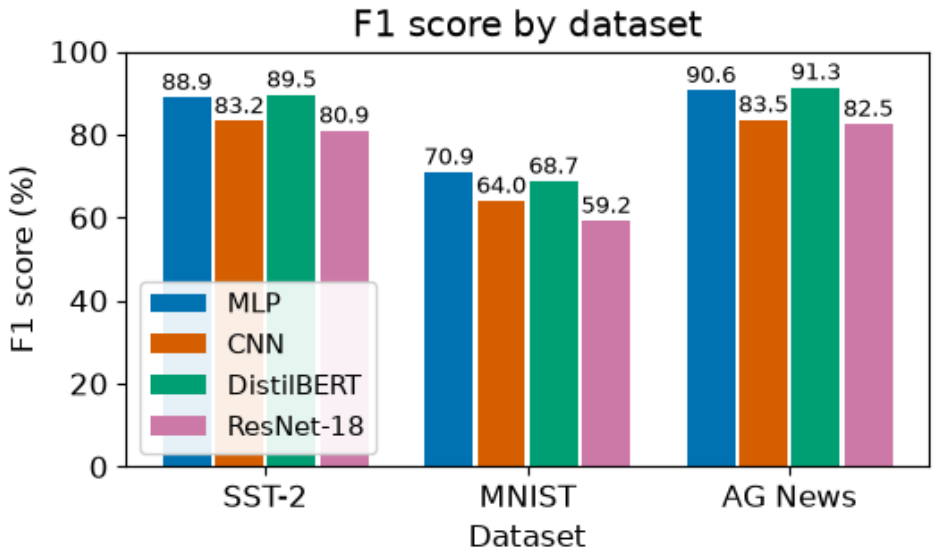


Figure 3: F1 score by dataset for the models in this experiment.

MLP outperforms CNN by 7 percentage points on MNIST. DistilBERT reaches 84.5% F1 score on SST-2. MLP outperforms ResNet-18 by 11.7 percentage points on MNIST (Figure 3).

Figure 4 shows the validation accuracy by dataset.

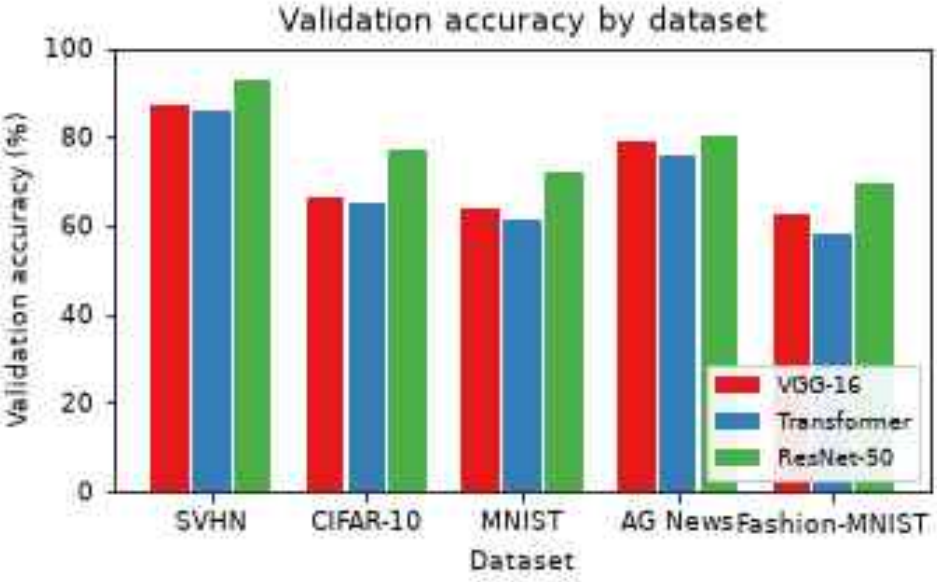


Figure 4: Validation accuracy by dataset for the models in this experiment.

ResNet-50 outperforms VGG-16 by 3.3 percentage points on AG News (Figure 4). As shown in Figure 4, VGG-16 achieves 64% validation accuracy on MNIST. ResNet-50 reaches 75.0% validation accuracy on CIFAR-10.

3 Discussion

Overall, the results are in line with what we expected from the literature.